

JUN LIU

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SUMMARY

First-year M.S. student in Computer Engineering at New York University, with solid experience in data structures and full-stack development.

Seeking a Software Engineer internship starting in May 2025

EDUCATION

The Ohio State University, Columbus

Ohio, U.S.A

Bachelor of Science in Computer Science & Engineering (GPA 3.5/4.0, Dean's List) | August 2020 - May 2024

New York University, Brooklyn

New York, U.S.A

Master of Science in Computer Engineering (Expected graduation: June 2026) | September 2024 - present

PUBLICATION

Jun Liu, "Application of Object Detection in Autonomous Driving," *2023 International Conference on Artificial Intelligence, Computers and Big Data (AICBD 2023)*.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, C, Go, Kotlin, PHP, HTML5, CSS

Frameworks & Libraries: Spring Boot, Express, Go Web frameworks, React, Angular, Android Jetpack, LangChain

Database: PostgreSQL, MySQL, Elasticsearch, Room, Firebase

WORK EXPERIENCE

Research Assistant

Ohio, U.S.A

Mentor: Dr. Alper Yilmaz, PCV Lab, The Ohio State University

May 2023 - December 2023

- Implemented an object detection system using the YOLOv5 model and DeepSORT algorithms to analyze stadium crowd in both pictures and videos, effectively tracking the crowd movements.
- Utilized homography to perform further analysis and detect anomalies, identifying individuals entering restricted areas.

PRACTICAL EXPERIENCE

OnlineOrder: A Spring Boot based online food ordering system

- Architected an online food ordering system using Spring Boot, with clear separation of structure into controllers, services, repositories and utilizing dependency injection to enhance maintainability and scalability.
- Constructed the frontend using React and CRUD REST APIs controllers, encompassing functionalities such as registration, menu searching, ordering, and check out. Deployed on AWS for efficient deployment.
- Developed Spring Data JDBC with PostgreSQL database and hosted on AWS RDS, handling data related to menus, shopping carts and restaurant operations.

SocialAI: An AI-Based social network

- Implemented a social network platform using React, focusing on user-friendly design and seamless interaction. Utilized OpenAI's Dall-E 3 to assist users in creating and update posts
- Used scalable web service in Go to handle posts and ElasticSearch (Deployed to Google App Engine) to provide search functionalities for users to search recent posts.
- Improved the authentication using token-based registration/login/logout flow with React Router v4 and server-side user authentication with JWT.

Next AI: Full-Stack Web-Based interactive AI Agent for PDF Document Queries

- Architected an interactive conversational UI leveraging React and Ant Design, enabling users to effortlessly upload and interact with PDF documents in real-time.
- Architected and implemented RESTful APIs via Express and Node.js, optimized for high performance request handling.
- Integrated OpenAI's GPT-3.5 Turbo API and LongChain technologies, achieving a sophisticated AI agent capable of document uploading, splitting, storage, retrieval, and output functionalities.

Mini Spotify: Mobile Music Application

- Designed a Kotlin Spotify app by fully leveraging Android Jetpack Library and Hilt Dependency Injection.
- Implemented the BottomBar & app Navigation using Jetpack Navigation component.
- Created RESTful APIs with json-server and used Retrofit to handle requests.
- Built the feed/album/favorite UI in Jetpack Compose following MVVM architecture and enabled the local cache ability for the favorite music feature by using Room Database.